Table 1: Local bilevel-refinement protocol used to generate the final refined gains.

Controller	**Tuned blocks**	**Frozen blocks**	**Local window**	**Objective summary**
DQ-MPCC	$\boldsymbol{q}_\phi$, $\boldsymbol{q}_c$, $\boldsymbol{q}_\ell$, $\boldsymbol{w}_u$, $\boldsymbol{q}_\omega$, Q_s	none	$[\times 0.75, \times 1.35]$ for $\boldsymbol{q}_\phi, \boldsymbol{q}_c, \boldsymbol{q}_\ell$; $[\times 0.92, \times 1.08]$ for $\boldsymbol{w}_u$; $[\times 0.90, \times 1.10]$ for $\boldsymbol{q}_\omega$; $[\times 0.75, \times 1.35]$ for Q_s	contour, lag, effective speed, progress, path ratio, effort, completion, solver failures
MPCC	$\boldsymbol{q}_q$, $\boldsymbol{q}_c$, $\boldsymbol{q}_\ell$, $\boldsymbol{w}_u$, $\boldsymbol{q}_\omega$, Q_s	none	$[\times 0.75, \times 1.35]$ for $\boldsymbol{q}_q, \boldsymbol{q}_c, \boldsymbol{q}_\ell$; $[\times 0.92, \times 1.08]$ for $\boldsymbol{w}_u$; $[\times 0.90, \times 1.10]$ for $\boldsymbol{q}_\omega$; $[\times 0.75, \times 1.35]$ for Q_s	contour, lag, effective speed, completion, solver failures

Combined Local Tuning Report

This standalone report documents the *local* bilevel refinement stage used to compare the manually stabilised operating points against the final refined gains for both controllers. It is intentionally self-contained for review: it states which gains were allowed to move, which gains were frozen by design, the multiplicative search windows, the final selected weights, and the corresponding closed-loop changes observed in the headless baseline simulations.

The key point is that this is not a global retuning from scratch. The refinement starts from the manual gains and searches only a bounded neighbourhood around them. In the current local-search setup, the effort and angular-rate penalty blocks are also allowed to move, so the reported gains correspond to a true local search over all cost-weight blocks listed in the tables below.

Each rollout in the tuning loop uses the same headless MiL configuration: maximum simulated duration $T_{\text{final}} = 30$ s, trajectory-geometry window $T_{\text{traj}} = 63$ s, control frequency $f_s = 100$ Hz ($T_s = 0.01$ s), prediction horizon $T_N = 0.3$ s discretised into $N = 30$ shooting nodes, and $N_w = 30$ arc-length interpolation waypoints. Under this setup, one rollout evaluates approximately 2971 closed-loop integration steps, and each Optuna trial evaluates one full rollout for every operating point in $\mathcal{V}$.

The DQ local study used 100 TPE trials over tuning velocities $\{5, 10, 16\}$ m/s, with mean trial time 53.4 s. The MPCC local study used 100 TPE trials over the same velocity set, with mean trial time 9.2 s.

Table 7: Closed-loop comparison between manual and refined gains used in the final baseline runs.

Controller	**Variant**	**RMSE$_c$**	**RMSE$_\ell$**	**v$_{\text{eff}}$**	**Path**	**Fail**
DQ-MPCC	Manual	1.791	1.813	2.738	100.00%	2.39%
DQ-MPCC	Refined	1.760	1.695	3.122	100.00%	2.35%
DQ-MPCC	Δ (%)	1.7	6.5	14.0	–	–
MPCC	Manual	1.825	1.890	3.898	100.00%	2.46%
MPCC	Refined	1.724	1.654	3.710	100.00%	2.02%
MPCC	Δ (%)	5.6	12.5	-4.8	–	–

Table 2: Exact upper-level objective coefficients used in the local refinement. Blank entries indicate terms not used by that controller.

Coefficient	**DQ-MPCC**	**Baseline MPCC**
W_{contour}	1.5	1.5
W_{lag}	0.35	0.35
$W_{\text{eff speed}}$	0.80	0.80
$W_{\text{incomplete}}$	1000	1000
W_{fail}	500	500
W_{progress}	–	–
$W_{\text{progress,hard}}$	–	–
Min progress ratio	–	–
$W_{\text{path ratio}}$	–	–
$W_{\text{path ratio,hard}}$	–	–
Min path ratio	–	–
W_{att}	–	–
W_{effort}	–	–

Table 3: Manual-reference metrics used to normalise the DQ-MPCC local objective at each tuning velocity.

$v_{\theta,\max}$	**RMSE**$_c$	**RMSE**$_\ell$	$\bar{v}_\theta$	Ratio	v_{eff}
5	0.299	0.384	1.594	0.793	1.264
10	0.346	0.507	1.594	0.685	1.092
16	0.346	0.507	1.594	0.685	1.092

Table 4: Manual-reference metrics used to normalise the Baseline MPCC local objective at each tuning velocity.

$v_{\theta,\max}$	**RMSE**$_c$	**RMSE**$_\ell$	$\bar{v}_\theta$	Ratio	v_{eff}
5	0.323	0.520	4.327	0.856	3.702
10	0.376	0.815	5.140	0.870	4.473
16	0.376	0.815	5.140	0.870	4.473

Table 5: DQ-MPCC local refinement: manual gains, multiplicative search window, and selected refined gains.

Weight	**Manual**	**Search interval**	**Refined**	**Status**
$\boldsymbol{q}_\phi$	$[0.65, 0.65, 1.9]$	$[0.49, 0.49, 1.4] \to [0.88, 0.88, 2.6]$	$[0.71, 0.86, 1.5]$	optimised
$\boldsymbol{q}_c$	$[40.0, 40.0, 40.0]$	$[30.0, 30.0, 30.0] \to [54.0, 54.0, 54.0]$	$[30.2, 32.7, 48.9]$	optimised
$\boldsymbol{q}_\ell$	$[3.0, 3.0, 3.0]$	$[2.2, 2.2, 2.2] \to [4.1, 4.1, 4.1]$	$[4.0, 3.2, 3.7]$	optimised
U^f	0.050	$[0.038, 0.068]$	0.046	optimised
$[U^{\tau_x}, U^{\tau_y}, U^{\tau_z}]$	$[70.0, 70.0, 70.0]$	$[52.5, 52.5, 52.5] \to [94.5, 94.5, 94.5]$	$[75.6, 65.2, 67.8]$	optimised
$\boldsymbol{q}_\omega$	$[0.020, 0.020, 0.10]$	$[0.015, 0.015, 0.075] \to [0.027, 0.027, 0.14]$	$[0.020, 0.019, 0.094]$	optimised
Q_s	11.0	$[8.2, 14.9]$	8.3	optimised
$J^\star_{\text{multi}}$		–	0.633	best local objective
Best trial		–	#95	Optuna index

Reviewer-facing interpretation. The two local studies should be read as refinements carried out under the same outer-level objective and the same velocity set, so the reported $J^\star_{\text{multi}}$ values are directly

Table 6: Baseline MPCC local refinement: manual gains, multiplicative search window, and selected refined gains.

Weight	**Manual**	**Search interval**	**Refined**	**Status**
$\boldsymbol{q}_q$	$[0.65, 0.65, 1.9]$	$[0.49, 0.49, 1.4] \rightarrow [0.88, 0.88, 2.6]$	$[0.62, 0.72, 1.6]$	optimised
$\boldsymbol{q}_c$	$[40.0, 40.0, 40.0]$	$[30.0, 30.0, 30.0] \rightarrow [54.0, 54.0, 54.0]$	$[40.6, 35.9, 33.7]$	optimised
$\boldsymbol{q}_\ell$	$[3.0, 3.0, 3.0]$	$[2.2, 2.2, 2.2] \rightarrow [4.1, 4.1, 4.1]$	$[4.0, 4.0, 2.7]$	optimised
U^f	0.050	$[0.038, 0.068]$	0.049	optimised
$[U^{\tau_x}, U^{\tau_y}, U^{\tau_z}]$	$[70.0, 70.0, 70.0]$	$[52.5, 52.5, 52.5] \rightarrow [94.5, 94.5, 94.5]$	$[70.9, 67.6, 72.2]$	optimised
$\boldsymbol{q}_\omega$	$[0.020, 0.020, 0.10]$	$[0.015, 0.015, 0.075] \rightarrow [0.027, 0.027, 0.14]$	$[0.020, 0.021, 0.095]$	optimised
Q_s	11.0	$[8.2, 14.9]$	8.3	optimised
$J^\star_{\text{multi}}$	–		0.863	best local objective
Best trial	–		#94	Optuna index

comparable as optimisation outcomes. The remaining differences arise from the controller formulations themselves and from the different closed-loop responses they induce under the same tuning protocol.

For DQ-MPCC, the refined gains reduce contouring RMSE by 1.7% and lag RMSE by 6.5%, while increasing effective speed by 14.0% and keeping both path completion and solver-failure ratio effectively unchanged. For the baseline MPCC, the refined gains reduce contouring RMSE by 5.6%, lag RMSE by 12.5%, and attitude RMSE by 6.4%, but reduce effective speed by 4.8%.

In the refreshed symmetric setup, the local search also includes the effort and angular-rate penalties. That makes the refined vectors more defensible for review, because the reported final gains are no longer conditioned on a partially frozen manual regularisation layer.